CSA02 – ASSIGNMENT 1

Grama Jyothi Rural Electricity Co-operative

TASK 3 – Define the Constants (3 Marks)

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3.1 Purpose of Constants

#define creates a preprocessor replacement and is useful for simple numeric values. const declares a typed object that cannot be modified through that identifier. enum defines named integer constants and is useful for household categories or status codes. Use uppercase names with underscores, such as RATE2_PAISE and FIXED_CHARGE_PAISE.

3.2 Complete the Code – Slab Billing

```
#include <stdio.h>

#define SLAB1_UNITS 100
#define SLAB2_UNITS 200
#define RATE1_PAISE 0
#define RATE2_PAISE 225
#define RATE3_PAISE 450
#define FIXED_CHARGE_PAISE 5000L

/* Returns the bill in paise. Negative units are outside this example's input assumptions. */
long compute_bill(int units)
{
    long paise = FIXED_CHARGE_PAISE;

    if (units > SLAB2_UNITS) {
        paise += (long)(units - SLAB2_UNITS) * RATE3_PAISE;
        paise += (long)(SLAB2_UNITS - SLAB1_UNITS) * RATE2_PAISE;
    } else if (units > SLAB1_UNITS) {
        paise += (long)(units - SLAB1_UNITS) * RATE2_PAISE;
    }

    /* The first slab costs RATE1_PAISE, which is zero. */
    return paise;
}

int main(void)
{
    int tests[] = {0, 100, 150, 200, 350};
    int n = (int)(sizeof tests / sizeof tests[0]);

    for (int i = 0; i < n; i++) {
        long p = compute_bill(tests[i]);
        printf("%3d units -> Rs %ld.%02ld\n",
               tests[i], p / 100, p % 100);
    }
    return 0;
}
```

Expected output

```
  0 units -> Rs 50.00
100 units -> Rs 50.00
150 units -> Rs 162.50
200 units -> Rs 275.00
350 units -> Rs 950.00
```

Compile and run

```
gcc -std=c11 -Wall -Wextra -o task03 task03.c
./task03
```

3.3 Tariff Revision Test

Change RATE2_PAISE from 225 to 230. The fixed charge remains 5000 paise and RATE3_PAISE remains 450 paise. Recalculate each bill:

Units	Original bill	Revised bill
0	■50.00	■50.00
100	■50.00	■50.00
150	■162.50	■165.00
200	■275.00	■280.00
350	■950.00	■955.00

Note: The corrected revised bill for 350 units is ■955.00: ■50 fixed charge + 100 × ■2.30 + 150 × ■4.50 = ■955.00. The 960.00 value in the source document is an arithmetic error.

3.4 Alternative Concept: const or enum

Typed const variables offer type checking and can be easier to inspect in a debugger. A const object may occupy storage, although optimisation can remove it. enum is suitable for named integer values, not fractional amounts. A #define normally does not create a variable or reserve storage for the constant itself. For these simple tariff values, macros are suitable.